

UChicago REU Notes Week 5

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1 Statements on G -categories

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To be honest, last week's note isn't as successful, so we'll mainly focus on the category theory from now on. Here, we assume that G is a finite group, and the categories (mentioned as objects of $\mathbf{Cat}$ or similar objects) are small.

1 Statements on G -categories

This part is devoted to understand some basics. I'll fill in the details for preliminaries on G -categories later, together with adjunction $(\mathbf{Ob} \dashv \mathcal{E})$ between $\mathbf{Cat}$ and $\mathbf{Set}$.

Given a category $\mathcal{A}$, the functor category $\mathbf{Cat}(\mathcal{E}(G), \mathcal{A})$ is automatically a G -category, as each element $g \in G$ can be realized as an automorphism functor on the indiscrete category $\mathcal{E}G$, so it admits both a left/right action by precomposing either g^{-1} or g as automorphism of $\mathcal{E}G$.

Similarly, if $\mathcal{A}$ itself already has a left/right G -action, then it defines a left/right G -action on $\mathbf{Cat}(\mathcal{B}, \mathcal{A})$ by postcomposition; similarly if $\mathcal{B}$ is a G -category, then precomposition also defines one. Combining the two, it admits a conjugation action on $\mathbf{Cat}(\mathcal{B}, \mathcal{A})$ if both $\mathcal{B}, \mathcal{A}$ admits a left action.

Given G -categories $\mathcal{A}, \mathcal{B}$, denote $\mathbf{Cat}_G(\mathcal{B}, \mathcal{A}) := \mathbf{Cat}(\mathcal{B}, \mathcal{A})$, and $G\mathbf{Cat}(\mathcal{B}, \mathcal{A}) := \mathbf{Cat}(\mathcal{B}, \mathcal{A})^G$ as the G -fixpoint (i.e. all functors commuting with G -actions).

Proposition 1.1. *There is a G -equivariant isomorphism when giving $\mathcal{A}, \mathcal{B}, \mathcal{C}$ three G -categories:*

$$\Phi : \mathbf{Cat}_G(\mathcal{A} \times \mathcal{B}, \mathcal{C}) \cong \mathbf{Cat}_G(\mathcal{A}, \mathbf{Cat}(\mathcal{B}, \mathcal{C}))$$

Proof. We recognize this as a non-equivariant statement first, then show if all three are G -categories (with G acts diagonally on $\mathcal{A} \times \mathcal{B}$), such isomorphism is G -equivariant.

I. Construction of Functor:

Given a functor $F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$, for any object $X \in \mathcal{A}$ (together with identity $\mathrm{id}_X : X \rightarrow X$), define a functor $F_X : \mathcal{B} \rightarrow \mathcal{C}$ by $F_X(Y) := F(X, Y)$, and given any morphism $g : Y \rightarrow Y'$ in $\mathcal{B}$, define the morphism $F_X g : F_X(Y) \rightarrow F_X(Y')$ as the morphism $F(\mathrm{id}_X, g) : F(X, Y) \rightarrow F(X, Y')$. The routine check that this is functorial, using the product category definition:

- Given $\mathrm{id}_Y : Y \rightarrow Y$, we have $F_X \mathrm{id}_Y = F(\mathrm{id}_X, \mathrm{id}_Y) = \mathrm{id}_{F(X, Y)} = \mathrm{id}_{F_X(Y)}$.

- Given $g : Y \rightarrow Y'$, $g' : Y' \rightarrow Y''$, we have

$$F_X(g' \circ g) = F(\text{id}_X, g' \circ g) = F(\text{id}_X, g') \circ F(\text{id}_X, g) = F_X g' \circ F_X g$$

Now, given a morphism $f : X \rightarrow X'$ in $\mathcal{A}$, we shall define a natural transformation $Nf : F_X \rightarrow F_{X'}$. Realistically, for any $Y \in \mathcal{B}$, the only choice is by $Nf_Y := F(f, \text{id}_Y) : F(X, Y) \rightarrow F(X', Y)$. Which, given $g : Y \rightarrow Y'$ in $\mathcal{B}$, the following diagram commutes using product category property:

$$\begin{array}{ccc} F(X, Y) & \xrightarrow{F(\text{id}_X, g)} & F(X, Y') \\ \downarrow F(f, \text{id}_Y) & & \downarrow F(f, \text{id}_{Y'}) \\ F(X', Y) & \xrightarrow{F(\text{id}_{X'}, g)} & F(X', Y') \end{array} \quad \equiv \quad \begin{array}{ccc} F_X(Y) & \xrightarrow{F_X g} & F_X(Y') \\ \downarrow Nf_Y & & \downarrow Nf_{Y'} \\ F_{X'}(Y) & \xrightarrow{F_{X'} g} & F_{X'}(Y') \end{array}$$

Hence, $Nf : F_X \rightarrow F_{X'}$ is a natural transformation.

Then, we claim that $N : \mathcal{A} \rightarrow \text{Cat}(\mathcal{B}, \mathcal{C})$ by $N(X) := F_X$, $Nf : F_X \rightarrow F_{X'}$ as above is a functor:

- $(N\text{id}_X)_Y = F(\text{id}_X, \text{id}_Y) = \text{id}_{F(X, Y)} = \text{id}_{F_X(Y)}$, so $N\text{id}_X = \text{id}_{F_X}$.
- Given $f : X \rightarrow X'$, $f' : X' \rightarrow X''$, then

$$N(f' \circ f)_Y = F(f' \circ f, \text{id}_Y) = F(f', \text{id}_Y) \circ F(f, \text{id}_Y) = Nf'_{Y'} \circ Nf_Y$$

So, $N(f' \circ f) = Nf' \circ Nf$.

Now, suppose two such functors $F, F' : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ has a natural transformation $\mu : F \rightarrow F'$, we want to construct a natural transformation $\bar{\mu} : N \rightarrow N'$, where $N, N' : \mathcal{A} \rightarrow \text{Cat}(\mathcal{B}, \mathcal{C})$ are functors associated to F, F' . Which, given any $X \in \mathcal{A}$, we see the collection $(\bar{\mu}_X)_Y := \mu_{(X, Y)}$ is a natural transformation $F_X \rightarrow F'_X$, due to the following diagram for all $g : Y \rightarrow Y'$ in $\mathcal{B}$:

$$\begin{array}{ccc} F(X, Y) & \xrightarrow{F(\text{id}_X, g)} & F(X, Y') \\ \downarrow \mu_{(X, Y)} & & \downarrow \mu_{(X, Y')} \\ F'(X, Y) & \xrightarrow{F'(\text{id}_X, g)} & F'(X, Y') \end{array} \quad \equiv \quad \begin{array}{ccc} F_X(Y) & \xrightarrow{F_X g} & F_X(Y') \\ \downarrow (\bar{\mu}_X)_Y & & \downarrow (\bar{\mu}_X)_{Y'} \\ F'_X(Y) & \xrightarrow{F'_X g} & F'_X(Y') \end{array}$$

This shows $\bar{\mu}_X : F_X \rightarrow F'_X$ is a natural transformation, or morphism within $\text{Cat}(\mathcal{B}, \mathcal{C})$. Then, notice that if $\mu : F \rightarrow F'$, $\mu' : F' \rightarrow F''$ are two natural transformations between functors $F, F', F'' : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$, the following holds:

$$(\overline{\mu' \circ \mu})_Y = (\mu' \circ \mu)_{(X, Y)} = \mu'_{(X, Y)} \circ \mu_{(X, Y)} = (\bar{\mu'}_X)_Y \circ (\bar{\mu}_X)_Y$$

This shows as natural transformations, $\bar{\mu}_X : N_X \rightarrow N'_X$ and $\bar{\mu'}_X : N'_X \rightarrow N''_X$ satisfies $(\overline{\mu' \circ \mu})_X = \bar{\mu'}_X \circ \bar{\mu}_X$, so $\overline{\mu' \circ \mu} = \bar{\mu'} \circ \bar{\mu}$ as natural transformations $\bar{\mu} : N \rightarrow N'$, $\bar{\mu'} : N' \rightarrow N''$.

So, define $\Phi(F : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}) := N : \mathcal{A} \rightarrow \text{Cat}(\mathcal{B}, \mathcal{C})$, and $\Phi(\mu : F \rightarrow F') := \bar{\mu} : N \rightarrow N'$ as above, it's a legitimate functor.

II. Isomorphism on Objects:

Suppose $\Phi(F) = \Phi(F')$, then it's clear F, F' have the same action on the objects (as all $X \in \mathcal{A}$ satisfies $F_X = F'_X$, so all $Y \in \mathcal{B}$ satisfies $F(X, Y) = F_X(Y) = F'_X(Y) = F'(X, Y)$). Now, given $f : X \rightarrow X'$ in $\mathcal{A}$ and $g : Y \rightarrow Y'$ in $\mathcal{B}$, we also see that:

$$F(f, g) = F(f, \text{id}_{Y'}) \circ F(\text{id}_X, g) = Nf_{Y'} \circ F_X g = F' f_{Y'} \circ F'_X g = F'(f, \text{id}_{Y'}) \circ F'(\text{id}_X, g) = F'(f, g)$$

($Nf = N'f$ is by $\Phi(F) = \Phi(F')$, then $F_X = F'_X$ also the same logic).

This shows $F = F'$, as its map on morphisms are also the same.

Conversely, given a functor $G : \mathcal{A} \rightarrow \mathbf{Cat}(\mathcal{B}, \mathcal{C})$, we'd like to construct a functor $\tilde{G} : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$. Object wise, define $G(X, Y) := [G(X)](Y)$ (here, $G(X) : \mathcal{B} \rightarrow \mathcal{C}$ is a functor), and given morphisms $f : X \rightarrow X'$ in $\mathcal{A}$, $g : Y \rightarrow Y'$ in $\mathcal{B}$, define morphism $G(f, g) : G(X, Y) \rightarrow G(X', Y')$ as the following morphism in the commutative diagram:

$$\begin{array}{ccc} G(X, Y) := [G(X)](Y) & \xrightarrow{G(X)g} & [G(X)](Y') \\ Gf_Y \downarrow & & \downarrow Gf_{Y'} \\ [G(X')](Y) & \xrightarrow{G(X')g} & [G(X')](Y') =: G(X', Y') \end{array}$$

Here, recall that for each $f : X \rightarrow X'$, $Gf : G(X) \rightarrow G(X')$ is a morphism in $\mathbf{Cat}(\mathcal{B}, \mathcal{C})$ (i.e. a natural transformation), so the above diagram holds.

Now, to verify functoriality, consider the following:

- Given id_X in $\mathcal{A}$, id_Y in $\mathcal{B}$, we get $G(\text{id}_X) = \text{id}_{G(X)}$, so each natural transformation $G(\text{id}_X)_Y = \text{id}_{[G(X)](Y)}$. Similarly, since $G(X)$ is a functor, we get $G(X)\text{id}_Y = \text{id}_{[G(X)](Y)}$. So, the diagram becomes:

$$\begin{array}{ccc} \tilde{G}(X, Y) := [G(X)](Y) & \xrightarrow{\text{id}_{[G(X)](Y)}} & [G(X)](Y) \\ \text{id}_{[G(X)](Y)} \downarrow & & \downarrow \text{id}_{[G(X)](Y)} \\ [G(X)](Y) & \xrightarrow{\text{id}_{[G(X)](Y)}} & [G(X)](Y) =: \tilde{G}(X, Y) \end{array}$$

Hence, $G(\text{id}_X, \text{id}_Y)$ results in the identity of $G(X, Y)$.

- Given $f : X \rightarrow X', f' : X' \rightarrow X''$ in $\mathcal{A}$, and $g : Y \rightarrow Y', g' : Y' \rightarrow Y''$ in $\mathcal{B}$, the following gigantic diagram holds, due to the functoriality $G(f' \circ f) = Gf' \circ Gf$, and each $G(X)(g' \circ g) = G(X)g' \circ G(X)g$:

$$\begin{array}{ccccccc} \tilde{G}(X, Y) := [G(X)](Y) & \xrightarrow{G(X)g} & [G(X)](Y') & \xrightarrow{G(X)g'} & [G(X)](Y'') & & \\ Gf_Y \downarrow & & Gf_{Y'} \downarrow & & Gf_{Y''} \downarrow & & \\ [G(X')](Y) & \xrightarrow{G(X')g} & \tilde{G}(X', Y') := [G(X')](Y') & \xrightarrow{G(X')g'} & [G(X')](Y'') & & \\ Gf'_Y \downarrow & & Gf'_{Y'} \downarrow & & Gf'_{Y''} \downarrow & & \\ [G(X'')](Y) & \xrightarrow{G(X'')g} & [G(X'')](Y') & \xrightarrow{G(X'')g'} & \tilde{G}(X'', Y'') := [G(X'')](Y'') & & \end{array}$$

Here, adjoining diagonal squares provide the composition $\tilde{G}(f', g') \circ \tilde{G}(f, g)$, while the edges of the largest square provides $\tilde{G}(f' \circ f, g' \circ g)$, the commutation shows the functoriality.

So, by routine check on the definition, we have $\Phi(\tilde{G}) = G$, showing surjectivity on the objects.

III. Naturality:

Let's skip this...

IV. Equivariance Version:

Assuming the first three statements, the remaining statement is to show: Φ as a functor commutes with G -action. Let $g \in G$, which given any $F \in \text{Cat}(\mathcal{A} \times \mathcal{B}, \mathcal{C})$, we have $g \cdot F := g \circ F \circ g^{-1}$ being the conjugation action of G on this functor category (where the left g is the functor on $\mathcal{C}$, and the right g^{-1} is the diagonal g^{-1} functor on $\mathcal{A} \times \mathcal{B}$). Let $N := \Phi(F)$, and $N' := \Phi(g \cdot F)$. Then, notice $g \cdot F$ satisfies the following, for all $f : X \rightarrow X'$ in $\mathcal{A}$, and $g : Y \rightarrow Y'$ in $\mathcal{B}$:

$$\begin{aligned} N'_X(Y) &= g \circ F \circ g^{-1}(X, Y), \quad (g \cdot N)_X(Y) = (g \circ N \circ g^{-1})_X(Y) = (g \circ N_{g^{-1}(X)})(Y) \\ &= g \circ N_{g^{-1}(X)}(g^{-1}(Y)) = g \circ F(g^{-1}(X), g^{-1}(Y)) = g \circ F \circ g^{-1}(X, Y) \end{aligned}$$

(Yeah, we also need to verify that they're identical on the morphisms of $\mathcal{B}$, but let's skip it here...)

Here, we do need to be careful on if g is a functor on the categories, or the functor on the functor categories (as the second $g \circ N$ is really g acting on the pointwise functor $N_{g^{-1}(X)}$ using conjugation).

This shows $N', g \cdot N$ are identical on the objects. Tp □

God the proof is actually long as hell...

Definition 1.2. Given a space X , the chaotic topological category $\mathcal{E}X$ has object space X , morphism space $X \times X$, such that:

- Identity map $i : X \rightarrow X \times X$ by $x \mapsto (x, x)$ is continuous (also the same as diagonal morphism).
- The source map $s : X \times X \rightarrow X$ by $s(x, y) := x$ is continuous (also the same as first projection).
- The target map $t : X \times X \rightarrow X$ by $t(x, y) := y$ is continuous (also the same as second projection).
- Composition $\circ : (X \times X) \times_{s,t} (X \times X) \rightarrow X \times X$ by $\circ((x, y), (y, z)) := (x, z)$ is continuous.

Since there is only one morphism between any two objects, a morphism is equivalent to specify its source and target.

In particular, for any $x \in X$, the maps $\eta_x, \mu_x : X \rightarrow X \times X$ by $\eta_x(y) := (x, y)$, and $\mu_x(y) := (y, x)$ are continuous (as they're fixing one coordinate, while doing inclusion on the right). These also form natural isomorphisms $\eta : x \rightarrow X$, and $\mu : X \rightarrow x$ as follow:

$$\begin{array}{ccc} x & \xrightarrow{(x,x)} & x \\ \eta_x(y)=(x,y) \downarrow & & \downarrow \eta_x(z)=(x,z) \\ y & \xrightarrow{(y,z)} & z \end{array}, \quad \begin{array}{ccc} y & \xrightarrow{(y,z)} & z \\ \mu_x(y)=(y,x) \downarrow & & \downarrow \mu_x(z)=(z,x) \\ x & \xrightarrow{(x,x)} & x \end{array}$$

This shows as a category, $\mathcal{E}X$ is equivalent to the terminal category $*$ (which only has one unique topological category structure).

Also, if $X = G$ is a topological group (including discrete ones!) then $\mathcal{E}G$ is isomorphic to the translation category of G (as all morphisms can be interpreted as a translation from one group element to the other).

Let Π be a topological group (intepreted as the category of single object $*$, with morphism space Π), then there is a group homomorphism $\Pi^{\text{op}} \rightarrow \text{Aut}_{\mathcal{T}\text{Cat}}(\mathcal{E}\Pi)$, by right multiplication. So, all $g \in \Pi$ satisfies $g : \mathcal{E}\Pi \rightarrow \mathcal{E}\Pi$ by $\sigma \mapsto \sigma g$ (which is a homeomorphism on Π by definition of topological group), and morphism $(\sigma, \tau) \mapsto (\sigma g, \tau g)$ (again, this is the continuous diagonal action on $\Pi \times \Pi = \text{Mor}(\mathcal{E}\Pi)$).

Proposition 1.3. *The orbit category $\mathcal{E}\Pi/\Pi$ is isomorphic to Π as topological category.*

Proof. It's clear on the object space, as $\text{Ob}(\mathcal{E}\Pi) = \Pi$ as set, so $\text{Ob}(\mathcal{E}\Pi)\Pi = *$.

The map on the morphism space is $(\sigma, \tau) \sim (\sigma', \tau')$ iff there exists $g \in \Pi$, such that $(\sigma, \tau) = (\sigma'g, \tau'g)$. Which, this is equivalent to say $\sigma\tau^{-1} = \sigma'gg^{-1}\tau'^{-1} = \sigma'\tau'^{-1}$. So, we see all $(\sigma, \tau) \sim (e, \tau\sigma^{-1})$, and all $g, g' \in \Pi$ satisfies $(e, g) \sim (e, g')$ iff $g^{-1} = g'^{-1}$, or $g = g'$. So, we see each (e, g) establishes a distinct equivalent class, hence providing a bijection between $(\Pi \times \Pi)/\Pi \cong \Pi$, by $(e, g) \mapsto g$.

Finally, to show this is an isomorphism as topological category, we see such identification of morphism is topologically isomorphic (because $\Pi \rightarrow \Pi \times \Pi \rightarrow \Pi$ by $g \mapsto (e, g) \mapsto g$ is a homeomorphism). Then, given (e, g) and (e, g') in the quotient, since the second one is equivalent to $(g, g'g)$, then we see the composition is $\circ((e, g), (g, g'g)) = (e, g'g)$, which has quotient $g'g$, which is equivalent to the composition of the image of (e, g') and (e, g) , showing it's preserving the composition map. Hence, it's an isomorphism of categories. $\square$

Theorem 1.4. *Given a G -space X , a topological group Π (regarded as a trivial G -space, by $G \rightarrow e \rightarrow \text{Aut}(\Pi)$), the quotient functor $p : \mathcal{E}\Pi \rightarrow \Pi$ induces an isomorphism if topological G -categories*

$$\text{Cat}(\mathcal{E}X, \mathcal{E}\Pi)/\Pi \xrightarrow{\sim} \text{Cat}_G(\mathcal{E}X, \Pi)$$

Proof. In terms of objects, we have $\text{Cat}(\mathcal{E}X, \mathcal{E}\Pi) \cong \text{Set}(X, \Pi)$ $\square$